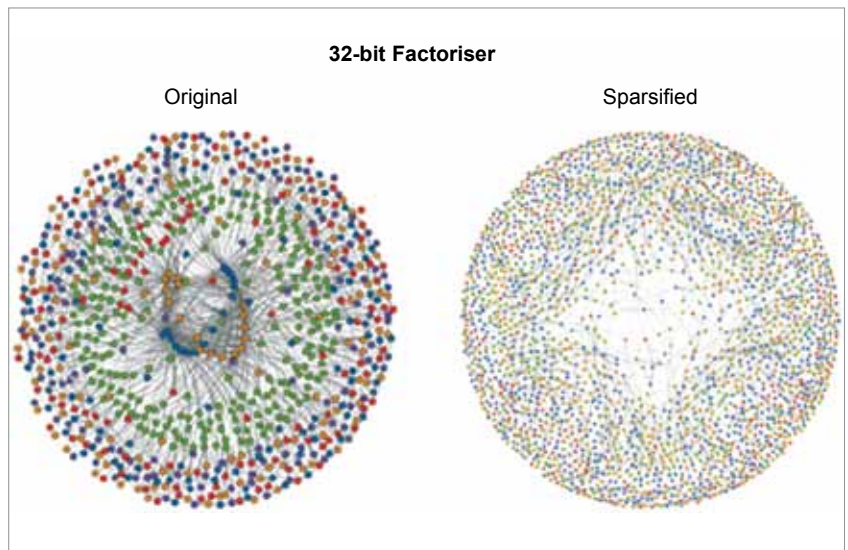




RESEARCH & INNOVATION UPDATES

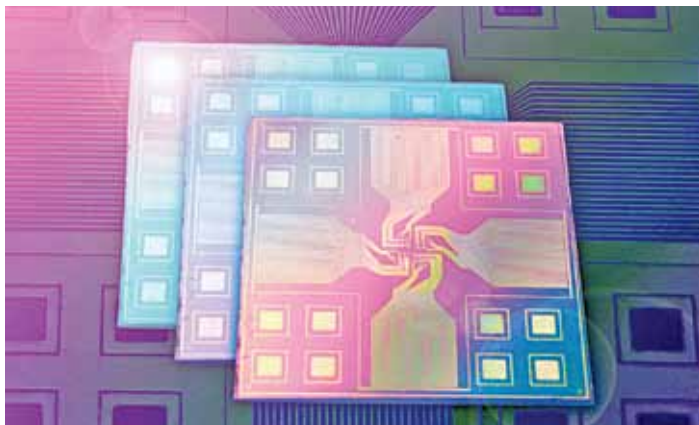
Will the new p-computers replace traditional computers?

It is a well-known fact that traditional computers are unable to compute in an energy-efficient manner. It is a restriction that has prompted a quest for new computing technologies. According to UC Santa Barbara researchers, probabilistic computers, or p-computers, are the answer. They created a one-of-a-kind sparse Ising machine (sIm), a revolutionary computing device that solves optimisation problems while consuming the least amount of energy. The sIm is a collection of probabilistic bits that oscillate between positions and work at normal temperatures. Their prototype was built on an FPGA, and they were able to prove that their design was up to six orders of magnitude quicker than traditional computers.



Hard optimisation problems can be expressed as interacting networks of probabilistic bits. Efficient solution of these problems require making them less dense at the expense of more p-bits (Credit: UC Santa Barbara)

MIT engineers build a LEGO-like AI chip



MIT engineers have created a reconfigurable AI chip that comprises alternating layers of sensing and processing elements that can communicate with each other (Credit: <https://news.mit.edu>)

With a LEGO-like design for a stackable, customisable artificial intelligence chip, MIT engineers have taken a step toward a modular chip design. Their system transmits data across the device using light rather than physical cables. The chip's layers communicate optically thanks to alternating layers of sensing and processing units, as well as light-emitting diodes (LEDs). As a result, the chip may be changed, with layers that can be swapped out or piled on to add new sensors or processors. Moreover, the chip employs artificial synapses—arrays of memory resistors, or memristors—that can be trained to process and classify signals directly on the chip, without an internet connection!